

19	D	We cannot use Coulomb's Law because that is for point charges (or charge distributions with spherical symmetry), but we have metal plates. Uniform electric field strength between the metal plates, $E = \frac{\Delta V}{d} = \frac{3V}{r_1 + r_2}$

		Electric force on charged oil drop = $qE = q\left(\frac{3V}{r_1 + r_2}\right)$ Distractors (remove from suggested solution for students): A → formula for electric potential energy if the plates were point charges B → formula for electric force if the plates were point charges C → calculated electric field strength due to each plate separately D → correct
20	B	work done by external force = $\Delta U = q\Delta V$